



CLOTH TRACER

Digital Hostel Laundry Management System

MANGLA LAUNDRY

React.js

Node.js

MySQL

Firebase Auth

Cloudinary

TEAM MATrix

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A complete working prototype | College Technical Project Presentation

THE PROBLEM: With Traditional Hostel Laundry

TRADITIONAL PROCESS

- 1 Student carries clothes to counter
- 2 Waits in queue for staff
- 3 Hand-written paper slip issued
- 4 Clothes washed off-record
- 5 Student returns hoping nothing is lost
- 6 Any issue -> verbal complaint only



Manual laundry submission



Long queues at the counter



Paper slips, hard to track



Lack of transparency



Missing / damaged clothes go unnoticed



Complaints handled manually



No easy laundry history for students

CLOTH TRACER- The Digital Solution

From paper slips to a fully digital, photo-verified, trackable laundry system — built and running end-to-end.

BEFORE

Manual laundry submission

Paper slips, easy to lose

No quantity-level tracking

No proof of what was submitted

Complaints handled verbally

AFTER — CLOTH TRACER



Digital laundry submission



Instant digital receipts



Clothing-wise quantity tracking



Photo-based laundry records



Structured complaint management

ALSO INCLUDED



Admin dashboard



Schedule management



Student notifications

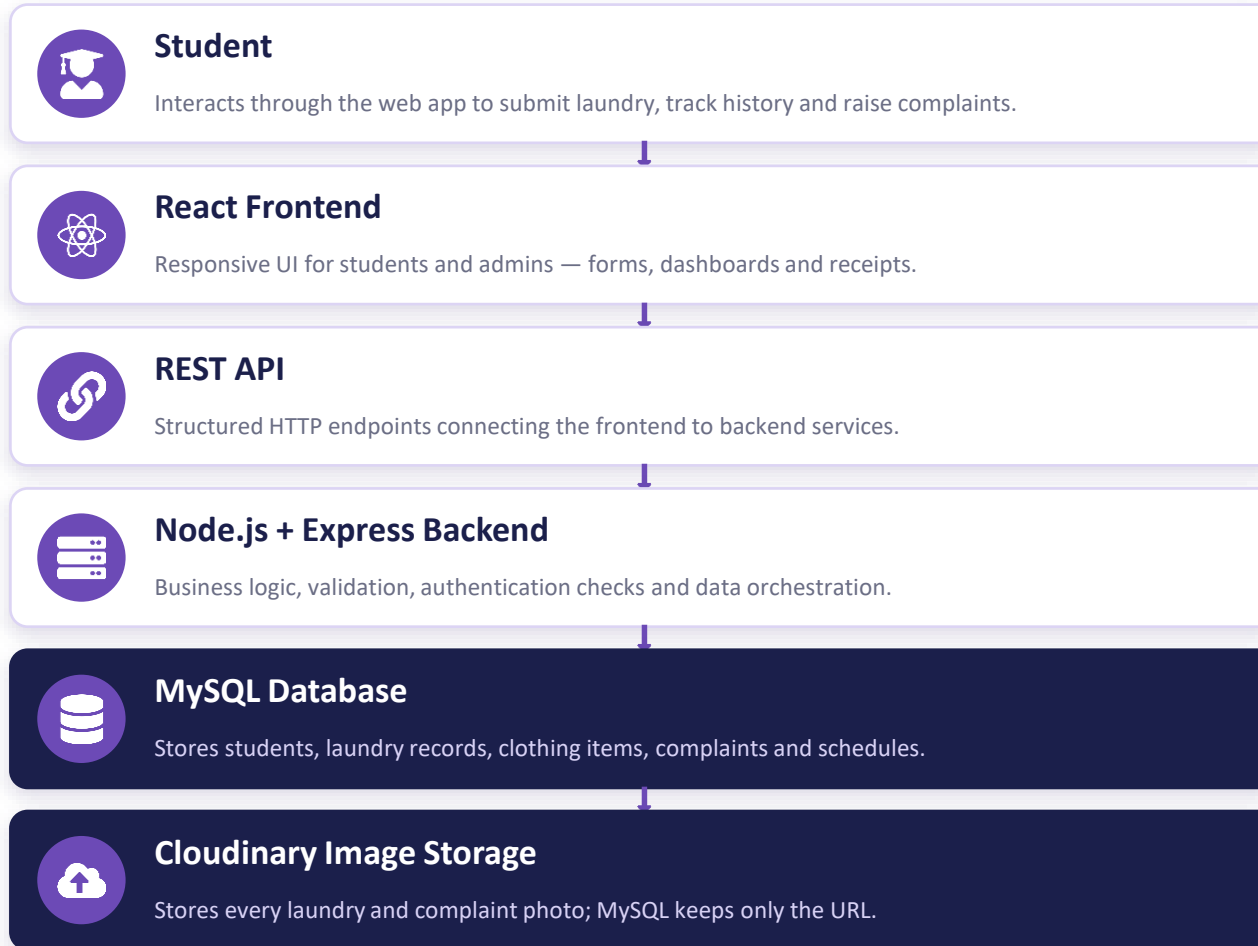


Full laundry history

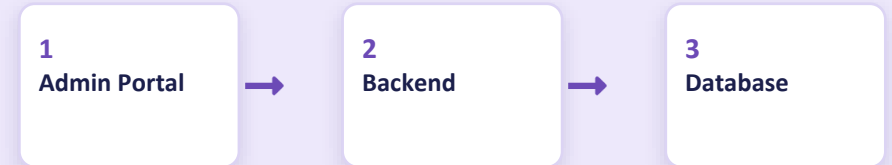


Centralized MySQL database

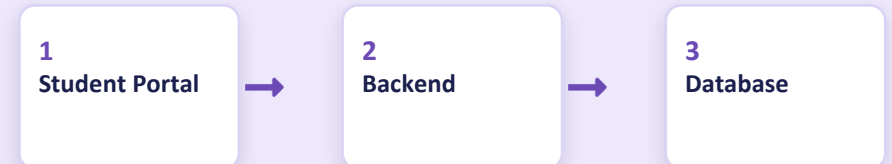
SYSTEM OVERVIEW:



ADMIN PORTAL FLOW



STUDENT PORTAL FLOW



Every layer talks only through the REST API — the frontend never touches MySQL or Cloudinary directly.

STUDENT SIDE:



Secure account access

- ✓ Authentication handled by Firebase Authentication
- ✓ Every student profile is stored in the MySQL database
- ✓ Editable profile information stays synchronized with the backend in real time

PROFILE FIELDS

Name

Roll Number

Email

Phone Number

Hostel

Admin-assigned Laundry Number

Edit Profile



FULL NAME

ROLL NUMBER

EMAIL ADDRESS

PHONE NUMBER

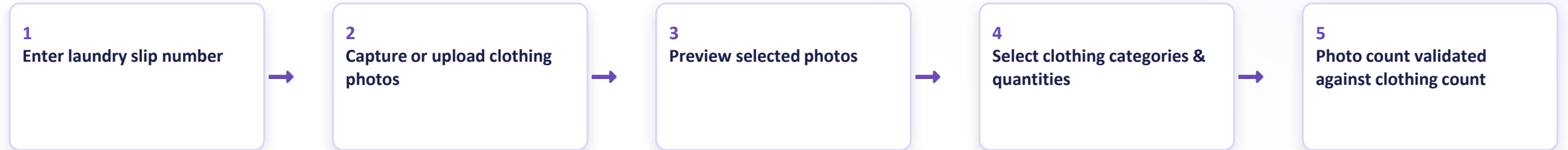
HOSTEL

LAUNDRY NUMBER (ADMIN-ASSIGNED)

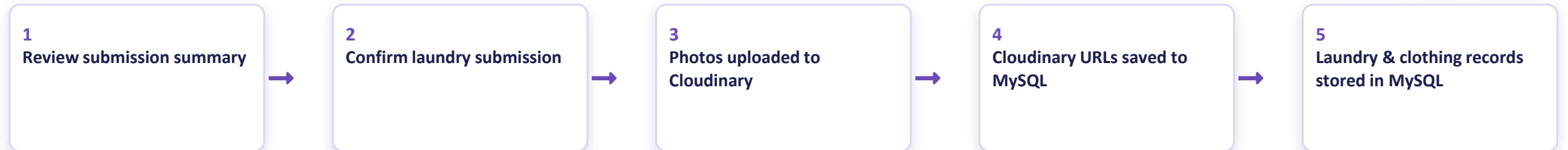
Save Changes

WORKFLOW- complete Laundry Submission

STAGE 1 — CAPTURE & VALIDATE



STAGE 2 — CONFIRM & STORE



Mobile-first submission flow — camera capture built directly into the web app.

Every submission is fully traceable: the exact photos taken at drop-off become permanent proof of what was submitted — used later for receipts and complaint verification.

Digital Receipt & Laundry History

- ✓ A digital receipt is generated instantly after every submission
- ✓ Receipt captures full laundry / slip details
- ✓ Clothing quantities are itemised and displayed
- ✓ Submitted photos are attached to the receipt
- ✓ Students can revisit their full laundry history anytime
- ✓ Admins can view every submitted laundry record centrally



DIGITAL RECEIPT

Slip #A2417 • Hostel G

T-shirt	2
Pant	1
Towel	2
Bed Sheet	1
Total Items	6






Attached photos

LAUNDRY HISTORY



Slip #A2417

Sep 05 · 6 items



Slip #A2390

Aug 29 · 4 items



Slip #A2355

Aug 22 · 7 items



Slip #A2310

Aug 15 · 3 items

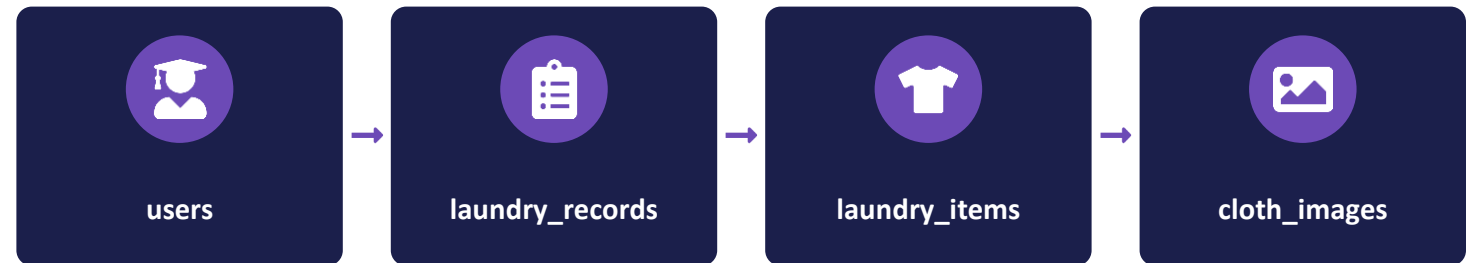
How Images & Data Connect



WHY CLOUDINARY, NOT MYSQL, FOR IMAGES

- ✓ MySQL stores structured information and image URLs only
- ✓ Cloudinary stores and serves the actual image files
- ✓ Keeps the relational database lightweight, fast and scalable

RELATIONSHIP AT A GLANCE



Complaint Management



ISSUE TYPES



No repeated proof uploads

Because every laundry photo is already stored against its record, the student never needs to re-upload evidence — the original submission photo becomes the complaint proof automatically.

Admin Portal

The admin portal is the central management system for the entire laundry operation — every student action surfaces here.



Dashboard



Laundry Management



Students



Hostels



Complaints



Reports



Settings



Notifications



Laundry Schedule



View History



Unidentified Clothes



Admin Access

Admin Dashboard & Schedule



842
Total Students



3,214
Laundry Submissions



96
Total Complaints



11
Open Complaints

TODAY'S SCHEDULE LOGIC

Monday	A · B · C · D · E · F · G
Tuesday	H · I · J · K · L · M · N
Wednesday	O · PG1 · PG2 · Q · FRG1 · FRG2
Thursday	A · B · C · D · E · F · G
Friday	H · I · J · K · L · M · N
Saturday	O · PG1 · PG2 · Q · FRG1 · FRG2
Sunday	OFF

RECENT ACTIVITY

Today's scheduled hostels
A, B, C, D, E, F, G

New complaint
Hostel L · Missing item

New complaint
Hostel B · Late delivery

Submissions today
58 laundry slips

The schedule is maintained in MySQL — the dashboard automatically resolves and displays the correct day.

Laundry Management

- ✓ View every laundry submission in one place
- ✓ Search by student name, roll number or slip number
- ✓ Filter records by hostel
- ✓ View clothing count and full item breakdown
- ✓ Inspect all submitted photos
- ✓ Open the full digital receipt for any record



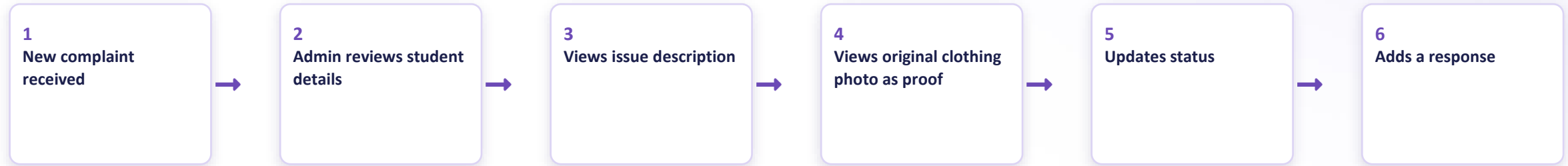
Search by name, roll no. or slip no.

Hostel: All ▾

Filter

Student	Slip #	Hostel	Items	Photos	Receipt
Aarav Sharma	A2417	G	6	4	View
Priya Nair	A2418	B	3	3	View
Rohit Verma	A2419	L	8	5	View
Simran Kaur	A2420	H	4	4	View
Devansh Rao	A2421	PG1	5	5	View

Complaint Resolution Workflow



COMPLAINT STATUS PIPELINE



Built for accountability and transparency

Every decision is grounded in the original photographic evidence from submission — nothing is resolved on word alone, and students always see exactly where their complaint stands.

Notifications & Reports



Notifications

SEND TO

All Students

Students by Hostel

Individual Student

NOTIFICATION TYPES



Laundry Reminder



Date Change



Laundry Ready



Reports



Total Students



Total Laundry Records



Total Clothes Processed



Complaint Statistics



Open Complaints



Under Review



Resolved Complaints

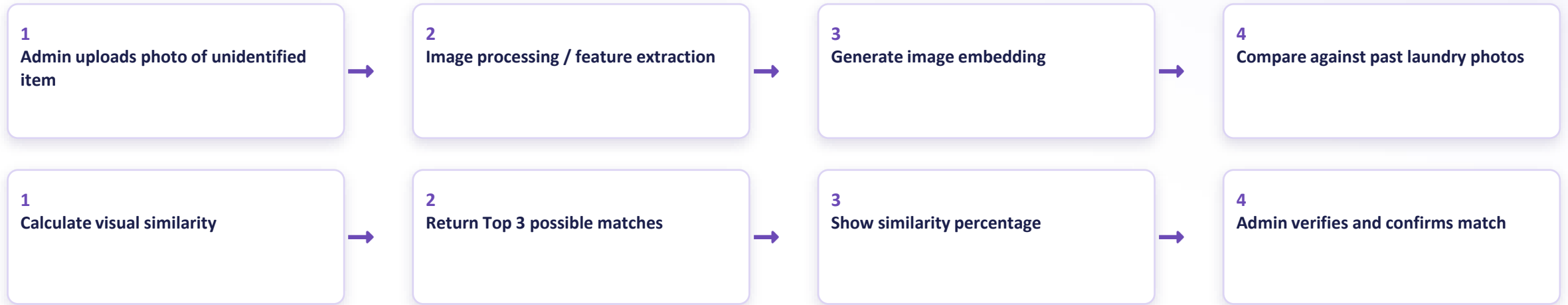


Hostel-wise Activity

These insights help administrators make faster, data-driven operational decisions.

Unidentified Cloth Matching (Proposed AI)

PROPOSED WORKFLOW



→ System identifies the likely student + laundry record → Student can be notified



Honest status of this feature

- ✓ This AI matching engine is a proposed future capability — it is not fully implemented in the current prototype.
- ✓ The prototype already stores every historical laundry photo and its database relationships, providing the data foundation this feature would need.
- ✓ Results would always be shown as a "Possible Match" with a similarity score — never a guaranteed identification — and require admin verification before action.

The Impact of CLOTH TRACER



For Students

- ✓ Less waiting in queues
- ✓ Complete digital records
- ✓ Photo-based accountability
- ✓ Easy, structured complaints
- ✓ Transparent status tracking
- ✓ Timely notifications



For Laundry Staff / Admin

- ✓ Centralized management
- ✓ Faster complaint resolution
- ✓ Better record keeping
- ✓ Full schedule visibility
- ✓ Searchable laundry history
- ✓ Reports and analytics

"CLOTH TRACER transforms hostel laundry from a manual, paper-based process into a transparent, accountable and digitally managed system."

THANK YOU

Questions?